DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III Topics in Analysis (3TN)

Tutorial Sheet 2 **1 August 2024**

1. Let (X, d) be a metric space. Let E, F be two subsets of X. Show if E is open and $E \cap F = \emptyset$, then $E \cap \overline{F} = \emptyset$, where $\overline{F}$ is the closure of the subset F.

2. Let (X, d) be a metric space. Let $E \subset X$.

   (a) Define what is meant by the statement that the set E is dense in X.

   (b) Show that if $f : X \to \mathbb{R}$ is a continuous function and E is dense in X,

$$\sup_{x \in X} f(x) = \sup_{x \in E} f(x) \quad \text{and} \quad \inf_{x \in X} f(x) = \inf_{x \in E} f(x)\,.$$

3. **(Uniform convergence and integration)**

   (a) Let (f_n) be a sequence of continuous functions from $[0, 1]$ to $\mathbb{R}$.

      (i) Define what is meant by the statement that (f_n) converges pointwise to f.

      (ii) Define what is meant by the statement that (f_n) converges uniformly to f.

      (iii) Show that if (f_n) converges uniformly to f then $\lim_{n\to\infty} \int_0^1 f_n(t)dt = \int_0^1 f(t)dt$.

      Hint: $\left| \int_0^1 f_n(t)dt - \int_0^1 f(t)dt \right| \leq \int_0^1 |f_n(t) - f(t)|dt$.

   (b) Consider the sequence $(f_n) : [0, 1] \to \mathbb{R}$ defined for every $x \in [0, 1]$ and every $n \in \mathbb{N}^*$ by

$$f_n(x) := \begin{cases} n^2 x(1 - nx) & \text{if } 0 \leq x \leq \dfrac{1}{n}\,, \\[2ex] 0 & \text{otherwise}\,. \end{cases}$$

      (i) Study the pointwise convergence of (f_n).

      (ii) Evaluate the integral $\int_0^1 f_n(x)\,dx$ and say whether the sequence (f_n) converges uniformly.

      (iii) Study the uniform convergence of (f_n) on the interval $[a, 1]$ with $a > 0$.

4. (a) Show that

$$\arctan(s + t) \leq \arctan s + \arctan t \qquad \forall s, t \geq 0\,.$$

   Hint: Consider $\dfrac{1}{1 + (s + \tau)^2} \leq \dfrac{1}{1 + \tau^2} \quad \forall \tau \geq 0$ and integrate over ????

(b) Let (X, d) be a metric space and let $d' : X \times X \to \mathbb{R}$ be defined by

$$d'(x, y) := \arctan(d(x, y)) \quad \forall x, y \in X .$$

(i) Show that d' is also a distance on X.
(ii) Say whether the distances d and d' are topologically equivalent.
(iii) Are they equivalent? Explain clearly your answer.

5. (a) Let A be a non-empty subset of $\mathbb{R}$. We recall that the diameter of A is

$$\operatorname{diam}(A) := \sup\{|x - y|\colon\ x, y \in A\} .$$

Show that

$$\operatorname{diam}(A) = \sup\ A - \inf\ A .$$

(b) Let $f : X \to \mathbb{R}$ be a bounded function. Deduce that

$$\sup_{x\in X} f(x) - \inf_{x\in X} f(x) = \sup\{|f(x) - f(y)|\colon\ x, y \in X\} .$$

6. Consider the sequence of functions (f_n) defined by

$$f_n(x) = \frac{nx}{nx + 1} \quad \forall x \in [0, 1], \quad \forall n \in \mathbb{N} .$$

(a) Study the pointwise convergence of the sequence (f_n).
(b) Does the sequence (f_n) converge uniformly? Explain your answer.
Hint: You can take $x_n = \dfrac{1}{n}$ for every $n \in \mathbb{N}^+$.

7. Consider the sequence (f_n) defined by $f_n(x) := \sin\left(\sqrt{x + 16n^2\pi^2}\right)$ for every $x \in [0, \infty)$ and for every $n \in \mathbb{N}$.

(a) Show that there exists $K > 0$ such that

$$|f_n(x) - f_n(y)| \leq K|x - y| \quad \forall x, y \geq 0, \ \forall n \in \mathbb{N} .$$

Hint: You can use $|\sin a - \sin b| \leq |a - b|$ $\forall a, b \in \mathbb{R}$ and $\sqrt{s} - \sqrt{t} = \dfrac{s - t}{\sqrt{s} + \sqrt{t}}$ $\forall s, t > 0$.

(b) Show that the sequence (f_n) converges pointwise to 0.
Hint: $\left(1 + \dfrac{x}{16n^2\pi^2}\right)^{1/2} \approx 1 + \dfrac{x}{32n^2\pi^2}$ for n large. (This comes from $\sqrt{1 + t} \approx 1 + \dfrac{t}{2}$ for $|t|$ small.)

(c) Consider the sequence of real numbers $x_n := \dfrac{\pi^2}{4} + 4n\pi^2$, use the hint in (b) to find an approximation of $f_n(x_n)$ and deduce that the sequence (f_n) does not converge uniformly.

8. Let (X, d_X) and (Y, d_Y) be two metric spaces. Let $f : X \to Y$ be a function. Show that

$$f \text{ continuous} \quad \Leftrightarrow \quad \forall E \subset X, \quad f(\overline{E}) \subset \overline{f(E)} .$$